# TEAM NAME: VECTOR CRAFT

TOPIC: POWER LINE AND TOWER INSPECTION DRONE

DESIGN REPORT


| Crew member | SC code | Role |
|---|---|---|
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- **Status of this report:** this is a design-stage report. Every number comes from hand calculations or from our own simplified Python model (Section 5). Nothing has been built, flown or measured. Part figures are planning values and we still have to check them against datasheets.

## 1. PROBLEM UNDERSTANDING AND ASSUMPTIONS

### MISSION:

- **What we have to do:** inspect high-voltage lines and towers from close range and report defects such as damaged insulators and loose fittings.
- **What the brief requires:** EMI-resistant avionics (compass, GPS, telemetry), stable hover in gusts up to 12 m/s, 3 m stand-off distance, 30 min endurance, 1.5 kg payload and obstacle avoidance.
- **The need behind it:** today people climb towers or use helicopters, which is slow and puts people near live equipment. A drone is only worth it if it gives a clear image of a fitting from a steady platform and never becomes a hazard itself. So safety near the conductor decides our design from the start, it is not an add-on.

**Table 1. How we read each requirement**

| Requirement | Our reading | Where we handle it |
|---|---|---|
| **EMI-resistant avionics** | Compass, GNSS and telemetry must work within a few meters of 50 Hz conductors and steel towers. | Section 3 (dual-antenna GNSS heading, CAN bus, shielding) |
| **Hover in 12 m/s gusts** | We hold position, not just attitude. Range to the nearest conductor must never drop under 3.0 m. | Sections 2 and 5 |
| **3 m stand-off** | A minimum distance. We fly 4.0 m nominal so a gust has 1.0 m of room. | Section 4 (LiDAR range loop, geofence) |
| **30 min endurance** | Hover and slow inspection with 20 % battery reserve left. | Section 2 (budget) |
| **1.5 kg payload** | RGB zoom camera, thermal camera, 2D LiDAR, gimbal and mounts, all inside 1.5 kg. | Section 2 mass budget |
| **Obstacle avoidance** | Detect wires, tower members and vegetation early enough to stop. | Section 4 |

### ASSUMPTIONS:

- **Air:** sea level, 1.225 kg/m$^3$. On a 40 °C day the air is about 8 % thinner and hover power goes up about 4 %. We keep that as a risk and did not size for it.
- **Wind:** worst case is wind blowing toward the line, because it pushes us at the conductor. We use a steady 8 m/s with a 1-cosine gust up to 12 m/s. No rain operation.
- **Stand-off:** 3.0 m is the hard minimum from the brief. The real safe distance depends on line voltage and the utility's rules, so it is a parameter in our software, not a fixed number.
- **Weight:** AUW is 5.68 kg with the 1.5 kg payload (26 % of AUW). 4.5 kg is not reachable with this payload and a 30 min battery (see Section 2).
- **Drag:** crosswind drag area CdA = 0.12 m$^2$, fitted so the simulated steady tilt at 8 m/s is 4.8°. This is an estimate, not a CFD result.
- **Efficiency:** hover figure of merit 0.65, motor and ESC efficiency 0.85, avionics and payload load 28 W, 80 % of battery energy usable. These stay assumptions until we have motor and propeller test data.
- **Rules:** the drone falls in the Small category (2 to 25 kg) of India's Drone Rules, 2021 [10].

## 2. CONCEPT SELECTION AND TRADE-OFFS

### ALTERNATIVES WE CONSIDERED:

We first threw out anything that cannot stop and hold 3 to 4 m from a conductor. That removes fixed-wing and wing-borne VTOL: they fly longer but cannot stay and look. That left five multirotor layouts.

**Table 2. Layout comparison**

| Layout | Good | Problem for this mission |
|---|---|---|
| **Large quadcopter** | Lightest and simplest | No motor-out margin. One failure and it falls, which is the worst case beside a live line. |
| **Flat hexacopter (chosen)** | Six motors share gust correction. Survives one motor out. | Slightly heavier than a quad. |
| **Flat octocopter** | Best redundancy | Two more motors, ESCs and arms (about 0.45 kg). By our model that cuts endurance by about 3 min, below 30 min. |
| **Coaxial X8** | Compact, redundant | Lower rotor efficiency, more parts and more vibration. |
| **Tilted-rotor hexacopter** | Resists side loads without tilting | Complex mixer and tuning, lower efficiency. We would rather spend effort on sensing and safety. |

## FINAL CHOICE: FLAT HEXACOPTER

- **Gusts:** six motors share the correction work, so each one only makes small thrust changes. In our gust run the motors stay at about 9.5 N of 19.6 N.
- **Motor-out:** with one motor failed the other five need at most 14.2 N each (73 % of 19.6 N) even during the gust. This is a static check.
- **Complexity:** it is less complicated and lighter than an octocopter, and it still has the redundancy we need.

## FRAME LAYOUT FOR GUST REJECTION:

- **Geometry:** flat hex with arms at 60° intervals, 1120 mm wheelbase (560 mm from the centre to each motor axis) and 18-inch (460 mm) counter-rotating propellers, 3 CW and 3 CCW. The gap between adjacent propeller tips is about 103 mm (560 mm minus 457 mm), so there is no tip interference.
- **Hub:** 3-plate carbon-fibre hub, 340 mm × 260 mm plates, with 92 mm lower and 70 mm upper standoff bays. Battery and electronics sit in the middle of the stack.
- **Arms:** Ǫ28/Ǫ25 mm tubes, CFRP inboard and GFRP outboard, joined with aluminium sleeves and clamp blocks. Motor shafts sit at Z = 128 mm.
- **Low frontal area:** one slim central pod, round tubes for the arms, and nothing sticking out sideways except two GNSS booms.
- **Centred mass:** battery and payload pod sit on the centreline, so the centre of gravity stays near the centre of thrust. A side gust then makes little unwanted pitching.

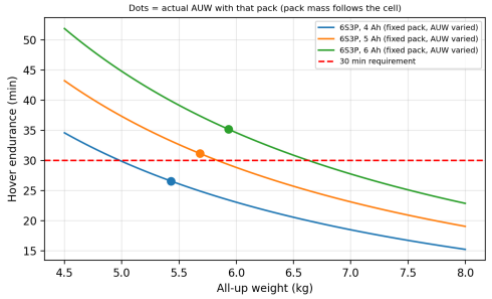



*Figure 2. Hover endurance against AUW for 4, 5 and 6 Ah cells. Dots show the real AUW with each pack.*

## SIZING, WEIGHT-POWER-ENDURANCE BUDGET:

- **Method:** we used momentum theory [1], with W = m·g and total disc area A:

$$P_{ideal} = W^{1.5} / \sqrt{2 \cdot \rho \cdot A} \qquad P_{battery} = P_{ideal} / (FM \cdot \eta) + P_{load} \qquad t = (E_{pack} \cdot 0.8) / P_{battery}$$

**Table 3. Mass budget (planning masses, to be replaced by CAD mass properties)**

| Item | kg | Note |
|---|---|---|
| Payload | 1.50 | RGB zoom camera, thermal camera, gimbal, 2D LiDAR, mounts |
| Frame (plates, arms, joints, mast, gear) | 1.05 | CFRP / GFRP tubes, aluminium joints |
| Motors x6 | 0.90 | About 150 g each, 2 kgf class |
| Propellers x6 | 0.12 | 18 inch |
| ESCs x6 | 0.18 | 30 A |
| Flight controller, GNSS x2, IMU, flow, radios, companion, wiring | 0.57 | |
| Battery 6S3P, 5 Ah cells | 1.36 | 18 cells at about 70 g plus 0.10 kg for strips, BMS and wrap |
| **Total AUW** | **5.68** | Payload is 26 % of AUW |

**Table 4. Sizing results**

| Quantity | Value | How |
|---|---|---|
| **Disc area / disc loading** | 0.99 $m^2$ / 57 $N/m^2$ | 6 × π $(0.457/2)^2$ ; W / A |
| **Ideal / shaft / battery power** | 268 / 412 / 513 W | FM 0.65, efficiency 0.85, plus 28 W load |
| **Hover current** | about 23 A | 513 W / 22.2 V (6 × about 9.3 N per motor) |
| **Pack energy / usable energy** | 333 / 266 Wh | 22.2 V × 15 Ah ; 80 % usable |
| **Hover endurance** | **31.2 min** | 266 Wh / 513 W |
| **Thrust-to-weight** | 2.11 | 6 × 19.6 N = 117.6 N against 55.7 N |
| **Peak battery current** | about 67 A | Our estimate at full thrust on all motors (about 11 A per ESC, well inside 40 A). The cells must handle about 22 A each. |

**Table 5. Cell capacity options (pack mass follows the cell, 6S3P)**

| Cell | Pack Wh | AUW kg | Hover W | Endurance | Thrust-to-weight |
|---|---|---|---|---|---|
| 4 Ah | 266 | 5.43 | 481 | 26.6 min | 2.21 |
| 5 Ah | 333 | 5.68 | 513 | 31.2 min | 2.11 |
| 6 Ah | 400 | 5.93 | 545 | 35.2 min | 2.02 |

- **Our pick:** 5 Ah cells in 6S3P as the baseline. 6 Ah cells would give about 35 min at 5.9 kg, but only if a cell with that capacity and mass can be bought. We have to confirm the real cell.
- **Correction to Day 1:** our Day 1 figure of 44.9 min used 100 % of the battery with no motor losses and no avionics load. Keeping a 20 % reserve takes it to 36 min, adding motor and ESC losses and the 28 W load takes it to 29 min, and the lighter 5.68 kg AUW brings it back to about 31 min. We use 31.2 min everywhere in this report.
- **What 44.9 min would need:** about 612 Wh of battery, which puts AUW at about 6.7 kg. That is above our 6 kg limit, so we are not designing for it.
- **Why not 4.5 kg:** payload 1.5 kg plus a 1.3 kg battery leaves only 1.7 kg for the frame, six motors, ESCs, propellers and avionics. We estimate those at 2.8 kg.

## 3. AVIONICS ARCHITECTURE

### OVERVIEW:

- **Split of jobs:** a Pixhawk-class flight controller running ArduPilot Copter handles everything safety-critical. A small companion computer handles LiDAR processing, stand-off logic and camera control. If the companion crashes, the flight controller keeps flying and its failsafes still work.

**Table 6. Avionics selection and placement (candidates, none bought or tested)**

| Function | Candidate | Placement | EMI measure |
|---|---|---|---|
| **Flight controller + IMU** | Holybro Pixhawk 6X , three IMUs on a vibration-isolated board | Damped plate at the centre of gravity | Metal case, shielded cables, away from ESC power leads |
| **Navigation 1: dual-antenna GNSS with RTK** | Dual-antenna receiver (Holybro H-RTK Unicore) | Two antennas on carbon booms either side of the mast, 0.8 m apart, above the top plate | Antennas away from power wiring, CAN link |
| **Navigation 2: optical flow** | Flow module with a downward rangefinder (for example HereFlow and TF02-Pro) | Bottom plate, facing straight down | Short cable, digital bus |
| **Navigation 3: industrial IMU** | extra external IMU | Rigid plate beside the flight controller, near the CG | Its magnetometer is turned off |
| **Obstacle and stand-off** | SLAMTEC RPLiDAR S2 | Front of the pod, scan plane vertical along the heading | Optical, shielded cable |
| **Payload** | Thermal core and 30x zoom GB camera on a 2-axis gimbal | Pod below the frame on rubber dampers, quick-release slide-in | Own filtered DC-DC supply, shielded video lines |
| **Companion computer** | Raspberry Pi 5 class, ROS 2 | Inside the hub, shielded | Own regulator, heartbeat to the flight controller |
| **Links** | 2.4 GHz digital link for control and video; sub-GHz telemetry and RC as backup (865-867 MHz band in India) | Antennas on the arms, away from GNSS | Two independent bands |
| **Power** | 6S3P Li-ion, current and voltage sensor, separate filtered 5 V supplies | Battery on the centreline | Star ground, twisted pairs, ferrites |

### NAVIGATION SENSOR PLACEMENT:

- **Dual-antenna GNSS with RTK:** near high-current lines the compass is the weakest sensor, so we take heading from the vector between two antennas 0.8 m apart. If the relative position error is 1 cm, heading error is about 0.7°. ArduPilot supports this as GPS for yaw . RTK gives centimeter-level position when corrections are available, so we plan to carry a portable base station. Steel towers cause multipath and can drop RTK, so our stand-off distance does not depend on RTK. It comes from the LiDAR.

- **Optical flow:** mounted on the bottom plate facing straight down. It only works within the range of its rangefinder over a textured surface, so it helps for take-off, landing and low-level moves. At tower height it cannot be the GPS fallback. There velocity comes from GNSS and the IMU, and range to the line from the LiDAR.
- **Industrial IMU:** the Pixhawk already has three IMUs. The external adds a fourth, higher-grade source so the EKF can reject a bad one. All sit on damped plates near the centre of gravity to avoid lever-arm errors.

**EMI APPROACH:**

- **Why we drop the compass for yaw:** the field of a long straight conductor is $B = \mu_0 I / (2\pi r)$ . At 3 m we get 13.3 µT at 200 A, 33.3 µT at 500 A and 66.7 µT at 1000 A. Earth's field is about 45 µT, so a heavily loaded line disturbs a magnetometer as much as the signal it measures. A balanced three-phase line partly cancels (about 44.2 µT at 1000 A with 4 m phase spacing, our assumption), and the field is 50 Hz AC with steel-tower distortion on top, so we treat it as an uncertainty, not a fixed offset.
- **Wiring:** power and signal wires are kept apart, GNSS and ESC cables are not routed together, sensor lines go through shielded tubes away from the power distribution board, and we use ferrites and filtering on payload and radio leads.
- **CAN bus:** GNSS and IMU use DroneCAN. Differential signalling rejects noise that couples into both wires the same way.
- **Links:** the 2.4 GHz link sits well above the frequencies where corona noise is strongest, and a second band is kept as backup.

**PAYLOAD SENSOR PLACEMENT:**

- **Pod:** we suspend the 1.5 kg payload under the main frame on a carbon-fibre plate with rubber dampers so motor vibration does not reach the cameras. It slides in and out with a quick-release for fast sensor swaps.
- **LiDAR:** at the front of the pod with the scan plane vertical and along the heading. A wire that runs left to right across our path then cuts that plane in one point and we read its range and height. A horizontal scan plane would run along the wire and often miss it. The pod hangs well below the rotor plane, so arms and propellers stay out of the horizontal view. The 90° blind sector of the 270° scan points down toward the landing gear.
- **Thin wires:** a 25 mm conductor at 3 m is only about 0.5° wide, so one scan at 0.25° resolution gives about 2 points, and fewer further out. Our software therefore confirms a wire over several scans. The single 2D plane also cannot see anything just above or below it, which we list as a risk.

## 4. SOFTWARE AND ALGORITHM DESIGN

**SOFTWARE SETUP:**

- **Flight controller:** ArduPilot Copter handles attitude and position loops, the EKF, geofence and failsafes. Our own work is parameters and failsafe settings.
- **Companion computer:** ROS 2 nodes for the LiDAR driver, line extraction, the stand-off controller, a health monitor and camera and gimbal control. It sends velocity commands and sends sector distances as obstacle distance messages so the flight controller's own avoidance can use them too.
- **Two safety rules:** the companion can only ask for motion, it cannot switch off a failsafe. And if its heartbeat stops, the flight controller treats it as a link loss.

**OBSTACLE AVOIDANCE ALGORITHM:**

1. LiDAR scans the surroundings:
   Measures distances to wires, tower members and other obstacles.
2. Filter the measurements:
   Removes unreliable or irrelevant readings, such as points too close to the drone or beyond 10 m.
3. Correct for drone tilt:
   Uses the drone's estimated roll and pitch to account for its orientation when interpreting the LiDAR points.
4. Identify and track obstacles:
   Groups nearby points into objects and checks whether those objects appear consistently across scans.
5. Send obstacle information:
   Reports the nearest confirmed obstacle’s distance and direction to the controller, which uses this information to help maintain a safe distance.

**STAND-OFF CONTROLLER:**

- **Control law:** it works along the line of sight using LiDAR range directly, so it still works if RTK drops. With d the measured range and d_ref = 4.0 m, the approach velocity is v = clip(Kp (d − d_ref), ±1.5 m/s) and the flight controller's velocity loop does the rest. We move along the line at 0.5 to 1 m/s and pause for each image.

**Table 7. Range bands around the conductor**

| Range | State | Behaviour |
|---|---|---|
| **3.6 m or more** | Normal | Hold 4.0 m, move along the line, capture images. |
| **3.3 to 3.6 m** | Amber | Stop moving along the line, correct the range, warn the pilot. |
| **Under 3.3 m** | Red | Retreat at up to 1.5 m/s until range is back above 3.6 m. |
| **3.0 m (geofence)** | Hard floor | A plane drawn 3.0 m from the surveyed line. A breach triggers brake and retreat. |

### AI CONTROL OF LIDAR, RGB AND THERMAL CAMERAS (DESIGN INTENT):

- **LiDAR:** the pipeline above produces the range and bearing that keep us at 4.0 m. It is the only sensor that controls stand-off.
- **Thermal camera:** the planned software looks for hot spots on joints and fittings by comparing each fitting's temperature with its neighbours and the ambient reading.
- **RGB zoom camera:** the planned software detects insulators and fittings, zooms to the target and checks for cracks, contamination or missing parts.
- **Camera control:** the gimbal centres on the detected target, and the image is only captured when we are inside the normal range band and the vehicle is steady. The thermal and RGB findings are combined into one report with position and time.
- **Status:** this is the planned design. The vision models have not been trained or tested on real line images, so we make no accuracy claims. The AI only controls the cameras and the report. It cannot override the failsafes or the range bands.

### FAIL-SAFE BEHAVIOUR NEAR LIVE CONDUCTORS:

- **Principle:** a normal return-to-launch in a straight line could cross the line, and landing in place could mean landing under a conductor or colliding with one. So we always move away from the conductor first, then decide, and we never land inside the danger zone. The drone records a backup return path on the way in, so it can reverse along a route that is already known to be clear.

**Table 8. Failsafe responses (thresholds are proposals to be tuned in simulation)**

| Trigger | Immediate action | Then |
|---|---|---|
| **Pilot override** | Manual control always has priority over any automatic mode. | Pilot flies on live video. Automatic retreat is on a switch. |
| **LiDAR invalid more than 0.3 s** | Stop along-line motion, hold on GNSS/IMU for up to 3 s. | Retreat along the recorded backup path to a staging point 10 m away. |
| **RTK lost** | Slow to 0.3 m/s, pause image capture. Range still comes from LiDAR. | If no fix in 30 s, retreat to staging. |
| **GNSS heading lost** | Hold yaw on the gyros, no compass fallback near lines. | Retreat to staging, then land at a safe zone. |
| **GNSS and LiDAR both lost** | Slow level back-away along the reverse heading for about 5 s. | Hand control to the pilot. No landing or loitering near the line. |
| **Control link lost** | Retreat along the recorded backup path, hold 10 s. | Return by a pre-planned corridor above the tower top plus margin. |
| **Battery 30 % / 20 %** | 30 %: finish the span, then retreat. 20 %: retreat now. | Fly the corridor home. 20 % is about 58 Wh, over 7 min of hover. |
| **Motor or ESC fault** | Retreat at once on the remaining five rotors. | Land at the nearest marked area 10 m from any conductor. |
| **Wind or tilt limit** | Tilt over 15° or range error over 0.8 m: abort and retreat. | Resume only if the wind estimate drops. |
| **EKF health alarm** | Stop, retreat, hand over to the pilot. | Land only at a marked safe zone. |

## Path Planning and Obstacle Avoidance

The proposed path-planning system helps the drone inspect transmission towers and power lines efficiently while avoiding obstacles and maintaining a minimum stand-off distance of 3 m.

### 1. Coverage Planning

The transmission network is represented as a graph, with towers as nodes and power-line sections as edges. A method such as Depth-First Search (DFS) can help plan the inspection sequence and reduce repeated visits.

### 2. A* Path Planning

The A* algorithm finds a suitable route from the drone's current position to its next inspection target. It considers the distance travelled and the estimated distance to the destination while avoiding known obstacles.

### 3. Dynamic Obstacle Avoidance

The drone uses LiDAR to detect nearby obstacles such as trees and tower structures. Camera-based detection may help identify birds. If an obstacle blocks the planned route, the system can update its map and calculate an alternative path.

### 4. Planning Workflow

The system first identifies the inspection targets and plans their visiting order. A* then calculates routes between targets, while sensor data helps detect obstacles and replan routes when needed. The process continues until all required sections are covered.

### 5. Design Considerations

The system combines coverage planning, route planning, and obstacle avoidance. Its performance depends on reliable sensing, accurate mapping, and maintaining safe clearance. Further simulation and testing are required before practical use.

## 5. SIMULATION METHODS AND RESULTS

### WHAT WE DID:

- **Tool:** a Python model (NumPy, SciPy, Matplotlib).
- **It covers:** the sizing in, a one-axis simulation of the stand-off loop in wind, a motor-out thrust check, the magnetic-field estimate and an arm bending check.

### MODEL AND INPUTS:

- **Stand-off model:** a point mass moving toward and away from the conductor. Tilt follows a second-order response ($\omega n = 9$ rad/s, $\zeta = 0.8$) to the tilt command and thrust rises to hold altitude ($T = W / \cos\theta$). Drag is $0.5 \cdot \rho \cdot CdA \cdot |v_{rel}| \cdot v_{rel}$ with the wind always blowing toward the line. The controller is a range loop from the LiDAR feeding a PI velocity loop.
- **Inputs:** AUW 5.68 kg, CdA 0.12 $m^2$, nominal range 4.0 m, LiDAR 40 Hz with 0.03 m noise and 50 ms latency, velocity noise 0.05 m/s, tilt limit 25°, max thrust 19.6 N per motor, time step 2.5 ms. Wind is a steady 8 m/s with small turbulence and a 1-cosine gust to 12 m/s over 6 s [11]. We start in equilibrium and ignore the first 20 s, so the results show the gust only.

**Table 9. Gust results (nominal 4.0 m, hard floor 3.0 m)**

| Case | Closest range | Excursion | Peak tilt | Peak motor | Result |
|---|---|---|---|---|---|
| **Ideal sensing** | 3.70 m | 0.30 m | 11.6° | 9.5 N | Above floor |
| **LiDAR noise and latency** | 3.68 m | 0.32 m | 11.7° | 9.5 N | Above floor |
| **Noise plus 0.5 s LiDAR dropout** | 3.69 m | 0.31 m | 12.0° | 9.5 N | Above floor |
| **Weak velocity loop (no integral)** | 2.57 m | 1.43 m | 11.8° | 9.5 N | **Breaks the floor** |

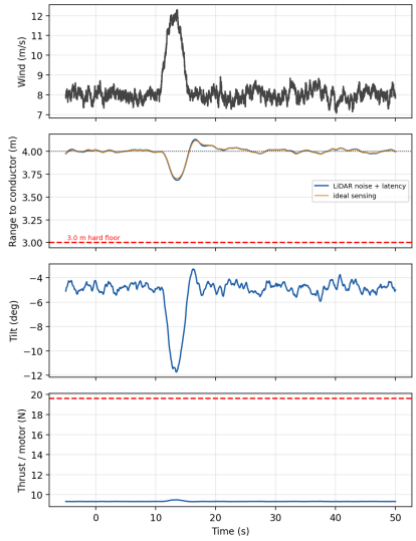



*Figure 3. Wind, range to conductor, tilt and motor thrust in the gust case.*

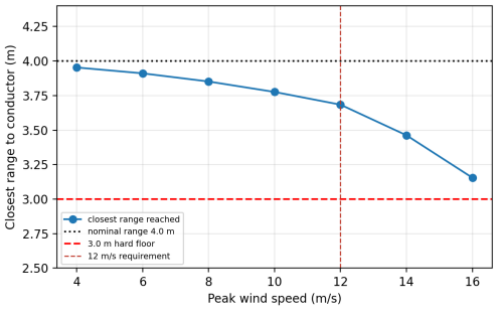



*Figure 4. Closest range against peak wind speed.*

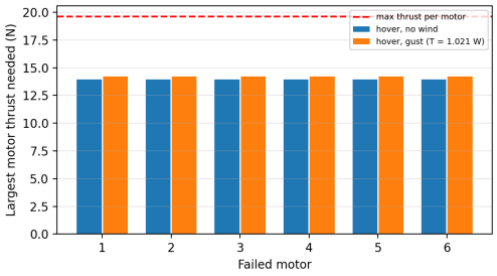



*Figure 5. Largest motor thrust needed with each motor failed.*

### WHAT THE RESULTS SHOW:

- **Gusts:** in the 12 m/s gust we get pushed about 0.32 m toward the line and tilt about 11.7°. The closest range is 3.68 m, above the 3.0 m floor. A 0.5 s LiDAR dropout costs almost nothing. The sweep (Figure 4) stays above 3 m up to 16 m/s (3.15 m).
- **Thrust is not the limit:** peak thrust is 9.5 N per motor of 19.6 N. What decides whether we hold the range is the sensing and the controller tuning. With a weak velocity loop the same gust takes us to 2.57 m. We will tune against ArduPilot SITL before we claim anything.
- **Why 4.0 m and not 3.0 m:** if we flew exactly 3.0 m, even a 0.3 m push toward the line would break the stand-off. The 1.0 m margin is what makes the result pass.
- **Motor-out:** the largest thrust needed is 13.9 N in hover (71 % of 19.6 N) and 14.2 N in the gust (73 %), with zero residual roll, pitch and yaw moment. Torque-to-thrust ratio is assumed as 0.02 m.

- **Arms:** at max thrust the CFRP root stress is 11.0 MPa and the GFRP stress at the joint is 6.0 MPa. With a 1.5 safety factor the tip deflection is 1.4 mm. Against our assumed allowables (200 and 150 MPa) the factors are above 12, so the arms look over-built and a lighter tube may be possible. The first bending frequency is about 44 Hz, which we have to compare with the rotor frequency at hover.

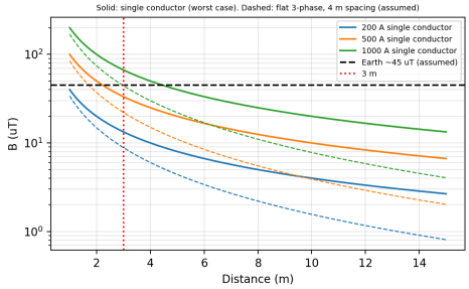



*Figure 6. Magnetic field against distance. Solid: single conductor (worst case). Dashed: flat three-phase line with 4 m spacing (assumed).*

**LIMITS OF THESE RESULTS:**

- The wind is uniform and always along the approach axis. Real gusts have side components, and the lattice of a tower makes its own wake. We have not modelled that.
- Rotor and ESC dynamics are collapsed into one attitude response, sensor noise is simple Gaussian noise, and altitude hold is ideal. Battery sag, cold and hot-day density are not modelled.
- Mass, efficiency, drag area, arm stiffness and allowable stresses are assumptions. They must be replaced with CAD, datasheet and test values. These results support the concept, they are not flight data.

## 6. RISKS AND NEXT STEPS

- **Thin endurance margin (about 1 min)**
- **Thin-wire detection by one 2D plane**
- **RTK dropouts and unproven EMI immunity:** stand-off comes from LiDAR, not GNSS. We will log GNSS signal quality, IMU noise and link quality near a live line before we trust the estimate.
- **Unproven software:** the failsafe logic and camera AI are designs only.

## 7. BOM

| component | specification | units | cost of 1 unit | total cost | PRICE TAKEN FROM |
|---|---|---|---|---|---|
| Brushless motors | T Motor Navigator MN4014 330KV | 6 | 8,392 | 50,352 | India Mart |
| propellors | 18 inches Carbon fiber | 3 pairs | | | |
| hexacopter frame | 3D printed | | | | |
| ESC | Hobbywing Xrotor 40 A | 6 | 2000 | 12000 | |
| Flight controller | Holybro Pixhawk 6C aluminium case variant | 1 | 35699 | 35699 | robu.in |
| GPS + IMU | CUAV C-RTK 2HP GNSS Heading Module | 1 | 25,887 | 25887 | Indian Robo Store |
| optical flow sensor | ARK flow MR | 1 | approx 30000 | 30000 | ARK Electronics |
| telemetry radio | Holybro SiK Telemetry Radio V3-100mW 433 Mhz | 1 | 9446 | 9446 | robu.in |
| battery pack | GenX Molicel+ 21.6V 13500mAh 6S 3P Premium Pack | 1 | 12900 | 12900 | Robokits India |
| power module distribution | Holybro PM07 14S Power Management Board | 1 | 5249 | 5249 | robu.in |
| LIDAR | SLAMTEC RPLIDAR S2 | 1 | 36379 | 36379 | robu.in |
| Thermal IR camera | Optris CS micro LT | 1 | 27000 | 27000 | Indiamart |
| payload plate and vibration isolators | | | | | |
| landing gear | | | | | |
| EMI protection and CAN wiring | Holybro Pixhawk 6C Cable Set | 1 | 2613 | 2613 | holybro store |
| ON board controller | Raspberry pi 5 model 8GB RAM | 1 | 19999 | 19999 | robu.in |
| RGB camera | SIYI A8 MINI 4K ULTRA HD 6X Digital zoom gimbal camera | 1 | 26999 | 26999 | Zbiotic |
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